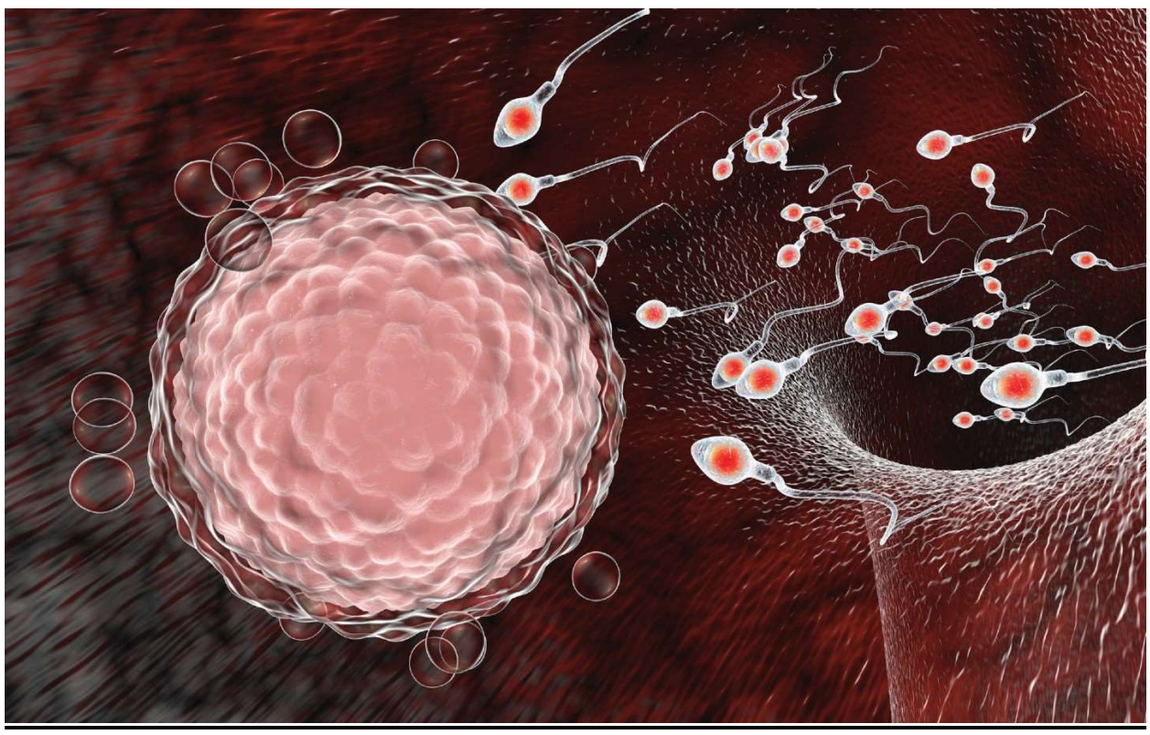
ZOO501- DEVELOPMENTAL BIOLOGY

Mid-Term File 2023 By

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Structure of Sperm?

The sperm has a head and tail. The head is oval-shaped and is only 10% of the overall length of the sperm. It contains the male genetic material in a structure called the nucleus. In addition to DNA, the sperm head also contains astructure called the acrosome.

Development biology:

Development biology is a discipline that studies embryonic and other developmental processes. It is the study of the process by which animals and plants grow and develop. Developmental biology also encompasses the biology of regeneration, asexual reproduction, metamorphosis, and the growth and differentiation of stem cells in the adult organism.

Cleavage is found in sea urchins:

Sea urchins exhibit radial holoblastic cleavage. The first and second cleavages are both meridional and are perpendicular to each other. That is to say, the cleavage furrows pass through the animal and vegetal poles. The third cleavage is equatorial, perpendicular to the first two cleavage planes, and separates the animal and vegetal hemispheres from one another. The fourth cleavage, however, is very different from the first three. The four cells of the animal tier divide meridionally into eight blastomeres, each with the same volume. These cells are called mesomeres. The vegetal tier, however, undergoes an unequal equatorial cleavage to produce four large cells, the macromeres, and four smaller micromeres at the vegetal pole.

What is different B/W Sex and Fertilization?

Fertilization: Fertilization is the process whereby two sex cells (gametes) fuse together to create a new individual with genetic potentials derived from both parents.

Sex involves the combining of genes from two different individuals into new arrangements.

Events of fertilization:

Fertilization consists of four major events:

1= Contact and recognition between sperm and egg. In most cases, this ensures that the sperm and egg are of the same species. 2= Regulation of sperm entry into the egg. Only one sperm can ultimately fertilize the egg. 3= Fusion of the genetic material of sperm and egg. 4= Activation of egg metabolism to start development.

Differenence between Animals pole and vegetal pole?

The region of the ovum which extrudes polar bodies and receives sperm is called animal pole, whereas the pole of the ovum opposite to the animal pole is known as vegetal pole.

Role of MPF:

MPF was first discovered as the major factor responsible for the resumption of meiotic cell divisions in the ovulated frog egg. It continues to play a role after fertilization, regulating the biphasic cell cycle of early blastomeres. Cleaving cells can be experimentally trapped in S phase by incubating them in an inhibitor of protein synthesis. When MPF is microinjected into these cells, they enter M. Their nuclear envelope breaks down and their chromatin condenses into chromosomes. After an hour, MPF is degraded and the chromosomes return to S phase. The primary function of MPF is

1= to promote spindle assembly, 2= chromatin condensation and 3= the breakdown of the nuclear envelope.

Vegetal Hemisphere:

The bottom half of the egg usually contains the yolk, it divides more slowly (because the large yolk deposits interfere with cleavage). This portion is the vegetal hemisphere of the egg. It is the region that is the lower, light-colored hemisphere of the egg, as its cells contain large amounts of dense yolk proteins.

What is microvilli:

Microfilaments are necessary for cell division, and they also are used to extend the egg surface into small projections that is called microvilli, which may aid sperm entry into the cell.

Diff between germ and somatic cell examples:

Germ Cell: In many species a specialized portion of egg cytoplasm gives rise to cells that are the Precursors of the gametes (the sperm and egg). The gametes and their precursor cells Are collectively called germ cells, and they are set aside for reproductive function. Example: Germ cells are egg cells in females and sperm cells in males.

somatic cell: All the other cells of the body are called Somatic cells. This separation of somatic cells (which give rise to the individual body) and germ cells (which contribute to the Formation of a new generation) is often one of the first differentiations to occur during Animal development. Examples: are muscle cells, blood cells, skin cells and nerve cells.

4 levels of Regulation of gene expression?

The regulation of gene expression can be accomplished at several levels:

- Differential gene transcription: It regulates that which of the nuclear genes is transcribed into RNA.
- Selective nuclear RNA processing: It regulating which of the transcribed RNAs (or which parts of such a nuclear RNA) enterinto the cytoplasm to become messenger RNAs.
- Selective messenger RNA translation: It regulates that which of the mRNAs in the cytoplasm becomes translated intoproteins.
- Differential protein modification: It regulates that which proteins are allowed to remain or function in the cell.

The two components of acrosomal reaction in sea urchins?

In most marine invertebrates, the acrosomal reaction has two components:

- The fusion Of the acrosomal vesicle with the sperm plasma membrane (an exocytosis that results In the release of the contents of the acrosomal vesicle).The extension of the acrosomal Process. The acrosomal reaction in sea urchins is initiated by contact of the sperm with The egg jelly.
- The second part of the acrosomal reaction involves the extension of the Acrosomal process. This protrusion arises through the polymerization of globular actin Molecules into actin filaments. The egg jelly factors that initiate the acrosomal reaction In sea urchins are often highly specific to each species.

What is naive?

An uncommitted cell can be described as naive, meaning that it has received no instructions directing it along a particular developmental pathway.

What is egg jelly?

Many types of eggs have glycoprotein meshwork called egg Jelly outside the vitelline envelope which is used either to attract or to activate sperm. The Egg, then, is a cell specialized for receiving sperm and initiating development.

The basic anatomy of eukaryotic gene:

The structure of eukaryotic genes. Most eukaryotic genes contain segments of coding sequences (exons) interrupted by noncoding sequences (introns). Both exons and introns are transcribed to yield a long primary RNA transcript.

Role of calcium in cortical reaction?

Role of calcium in cortical reaction Upon fertilization, the intracellular calcium ion concentration of the egg increases greatly. In this high-calcium environment, the cortical granule membranes fuse with the egg plasma membrane, releasing their contents. Once the fusion of the cortical granules begins near the point of sperm entry, a wave of cortical granule exocytosis propagates around the cortex to the opposite side of the egg. Several experiments have demonstrated that calcium ions are directly responsible for propagating the cortical granule reaction, and that these calcium ions are stored within the egg itself.

Classes of cadherin:

1= E-cadherin: a. Epithelial cadherin is expressed on all early mammalian embryonic cells, even at the 1cell stage. Later, this molecule is restricted to epithelial tissues of embryos and adults.

2= P-cadherin: a. Placental cadherin appears to be expressed primarily on the trophoblast cells (those placental cells of the mammalian embryo that contact the uterine wall) and on the uterine wall epithelium.

3= N-cadherin: a. Neural cadherin is first seen on mesodermal cells in the gastrulating embryo as they lose their E-cadherin expression. It is also highly expressed on the cells of the developing central nervous system.

4= Protocadherins: a. Calcium-dependent adhesion proteins that differ from the classic Cadherins in that they lack connections to the cytoskeleton through catenins. Protocadherins have been found to be very important in separating the notochord from the other mesodermal tissues during Xenopus gastrulation.

How cleavage cell can be trapped?

Cleaving cells can be experimentally trapped in S phase by incubating them in an inhibitor of protein synthesis. When MPF is microinjected into these cells, they enter M. Their nuclear

envelope breaks down and their chromatin condenses into chromosomes. After an hour, MPF is degraded and the chromosomes return to S phase.

What is developmental hierarchy?

The developmental hierarchy is the hierarchical series of decisions involving specific preexisting cell types.These developmental decisions are usually irreversible, that progressively and irreversibly restrict cell fate. Animal development is the division of embryo into the three germ layers.

1. Ectoderm: The ectoderm is one of the three primary germ layers formed in early embryonic development. It is the outermost layer, and is superficial to the mesoderm (the middle layer) and endoderm (the innermost layer).

2. Mesoderm: The mesoderm is a germ layer that arises during gastrulation, and is present between the ectoderm, which will turn into skin and central nervous system cells, and the endoderm, which will produce the gut and the lungs.

3. Endoderm: Endoderm is one of the germ layers—aggregates of cells that organize early during embryonic life and from which all organs and tissues develop.

Write a note on late responses in the Activation of sperm Egg Metabolism:

Late responses:

Shortly after the calcium ion levels rise in a sea urchin egg, its intracellular pH also increases. The rise in intracellular pH begins with a second influx of sodium ions, which causes a 1:1 exchange between sodium ions from the seawater and hydrogen ions from the egg. This loss of hydrogen ions causes the pH to rise. It is thought that the pH increase and the calcium ion elevation act together to stimulate new protein synthesis and DNA synthesis. The late responses of fertilization brought about by these ionic changes include the activation of DNA synthesis and protein synthesis.

Role of estrogen in egg development?

Estrogen is a hormone that can instruct the liver to make and secrete the yolk proteins, which are then transported through the blood into the enlarging eggs in the ovary. The yolk is transported into the bottom portion of the egg. Is also help to stimulate the growth of the egg follicle; they also stimulate the pituitary gland in the brain to release hormones that assist in follicular development.

What is cortex in egg?

The egg cortex is a large assemblage of the plasma membrane, extracellular coat, and subsurface skeleton to which a characteristic set of cytoskeletal elements, or- ganelles and macromolecules adhere. It behaves as a func- tional unit during the reorganizations that precede and follow fertilization.

Enlist developmental stages of frog?

Basic Facts. Frogs have multiple stages in their life cycle: egg, tadpole, tadpole with Legs, froglet, frog.

Stage 1: Egg. A frog begins life as a fertilized egg.

Stage 2: Tadpole. When the tadpole hatches, it looks more like a fish than a frog.

Stage 3: Young Frog. The tadpole grows two front legs and its long tail becomes shorter and shorter.

Stage 4: Adult Frog.

What causes this cyclic activity of MPF?

Mitosis-promoting factor contains two subunits.

1= The large subunit is called cyclin B.

2= Small subunit of MPF, the cyclin-dependent kinase

Cyclin B is often encoded by mRNAs stored in the oocyte cytoplasm, and if the translation of this message is specifically inhibited, the cell will not enter mitosis. The presence of cyclin B depends upon its synthesis and its degradation. Cyclin B regulates the small subunit of MPF, the cyclin-dependent kinase.

Core composition of Axoneme:

The axoneme consists of microtubule doublets (MTDs) and many other proteins encapsulated in the plasma membrane. The axoneme from most of the species has a 5- to 10-μm length and an ~300-nm diameter. Figure 1. Axoneme structure from various species and locations of cilia.

Bicoid and nanos:

In Drosophila, for instance, the anterior most portion of the egg contains an mRNA that encodes a protein called Bicoid.

The posterior most portion of the egg contains an mRNA that encodes a protein called Nanos.

- The Bicoid and Nanos proteins form a coordinate system based on their ratios, such that each region of the embryo will be distinguished by a different ratio of the two proteins.
- Those nuclei in regions containing high amounts of Bicoid and little Nanos will be instructed to activate those genes necessary for producing the head.
- Those nuclei in regions with slightly less Bicoid but with a small amount of Nanos will be instructed to activate those genes that generate the thorax.
- Those nuclei in regions that have little or no Bicoid and plenty of Nanos will be instructed to form the abdominal Structures.

Five basic steps of interaction of sperm and egg.

The interaction of sperm and egg generally proceeds according to five basic step:

1= The chemo-attraction of the sperm to the egg by soluble molecules secreted by the egg. 2= The exocytosis of the acrosomal vesicle to release its enzymes. 3= The binding of the sperm to the extracellular envelope (vitelline layer or zona pellucida) of the egg. 4= The passing of the sperm through this extracellular envelope. 5= Fusion of egg and sperm cell plasma membranes.

What are the function of poly (tail) mRNA.

1= The poly(A) tail promotes degradation of the mRNA. 2= The poly(A) tail Confers stability on the messenger RNA. 3= Allows the mRNA to exit the nucleus. Permits the mRNA to be translated into protein.

What are the function of Leeuwenhoek:

Leeuwenhoek is universally acknowledged as the father of microbiology. He discovered both protists and bacteria [1]. More than being the first to see this unimagined world of 'animalcules', he was the first even to think of looking—certainly, the first with the power to see.

J.Hammerling Hypothesis?

J.Hammerling exchanged nuclei between two morphologically distinct species, A. mediterranea and A. crenulata. J. Hammerling found that when he transferred the nucleus from one species into the stalk of another species, the newly formed cap eventually assumed the form associated with the donor nucleus. Thus, the nucleus was seen to control Acetabularia development. One hypothesis proposed on the base of Hammerling study is that, “The nucleus synthesizes a stable mRNA that lies dormant in the cytoplasm until the time of cap formation”. Hence, the expression of the cap is controlled not only by nuclear transcription, but also by the translation of the cytoplasmic RNA. In this unicellular organism, “development” is controlled at both the transcriptional and translational levels.

Write the attractive hormone:

Testosterone and estrogen drive lust; dopamine, norepinephrine, and serotonin create attraction; and oxytocin and vasopressin mediate attachment.

Pax6:

Paired box gene 6. The Pax6 transcription factor, which is needed for mammalian eye, Nervous system, and pancreas development, contains two potential DNA-binding domains.

Role of enhancer and promoter difference:

Promoter: A region, which is responsible for the binding of RNA polymerase and for the subsequent initiation of transcription.

Enhancer: It is a DNA sequence that can activate the utilization of a promoter, controlling the efficiency and rate of transcription from that particular promoter. The human β-globin gene has an enhancer in its 3′ UTR, roughly 700 base pairs.

Parts of egg:

Anatomy of an Egg The egg is composed of many parts, namely:

1= Yolk 2= Albumen or egg white 3= Shell membranes 4=Shell

The Shell: The shell is the egg's outer covering, accounting for about 9 to 12% of its tota weight depending on egg size. The shell is the egg's first line of defense against bacterial contamination.

The shell membrane: The Shell Membranes Lying between the eggshell and albumen, these two transparent protein membranes provide efficient defense against bacterial invasion.

Yolk: Yolk color ranges from just a hint of yellow to a magnificent deep orange, according to the feed and breed of the hen. The yolk has important nutritional and functional qualities.

Albumen: The egg white is known as the albumen, which comes from albus, the Latin word for “white.” Albumen accounts for most of an egg's liquid weight, about 67%. It contains more than half the egg's total protein.

Function of poly A tail:

Polyadenylation is the addition of a poly(A) tail to an RNA transcript, typically a messenger RNA (mRNA). The poly(A) tail consists of multiple adenosine monophosphates; in other words, it is a stretch of RNA that has only adenine bases. ... In many bacteria,

- The poly(A) tail promotes degradation of the mRNA.
- The poly(A) tail Confers stability on the messenger RNA.
- Allows the mRNA to exit the nucleus. Permits the mRNA to be translated into protein.

What is Air space?

An air space forms when the contents of the egg cool and contract after the egg is laid. The air cell usually rests between the outer and inner membranes at the egg’s larger end.

Conditional specification?

1= It is characteristic of all vertebrates and few invertebrates. 2= Specification by interactions between cells. 3= Relative positions are important. 4= Variable cleavages produce no invariant fate assignments to cells. 5= Massive cell rearrangements and migrations precede or accompany specification. 6= Capacity for "regulative" development: allows cells to acquire different functions.

How sea urchin avoided the polyspemy:

Species have evolved ways to prevent the union of more than two haploid nuclei. The most common way is to prevent the entry of more than one sperminto the egg. The sea urchin egg has two mechanisms to avoid polyspermy:

- A fast reaction, accomplished by an electric change in the egg plasma membrane.
- A slower reaction, caused by the exocytosis of the cortical granules.

Difference between mono and polyspermy?

In normal monospermy, in which only one sperm enters the egg, a haploid sperm nucleus and a haploid egg nucleus combine to form the diploid nucleus of the fertilized egg (zygote).

The entrance of multiple sperm known as **polyspermy** leads to disastrous consequences in most organisms.

Write the examples of transcription factors:

The transcription factors Oct4, Sox2, Klf4 and Nanog act as triggers for the induction of somatic cells to pluripotent stem cells.

Write down the protective chemicals in egg:

Many eggs contain ultraviolet filters and DNA repair enzymes that protect them from sunlight; some eggs contain molecules that potential predators find distasteful; and the yolk of bird eggs even contains antibodies. Many types of eggs have glycoprotein meshwork called egg jelly outside the vitelline envelope which is used either to attract or to activate sperm. The egg, then, is a cell specialized for receiving sperm and initiating development.

Define organogenesis in terms of germ layers:

Organogenesis is the process by which the three germ tissue layers of the embryo, which are the ectoderm, endoderm, and mesoderm, develop into the internal organs of the organism.

Explain cytokinesis and karyokinesis:

Cytokinesis is the physical process of cell division, which divides the cytoplasm of a parental cell into two daughter cells. It occurs concurrently with two types of nuclear division called mitosis and meiosis, which occur in animal cells.

Karyokinesis: During mitosis or meiosis, the process of karyokinesis is the division of cell nucleus. It is succeeded by cytokinesis, which is the splitting of the cytoplasm and cell membrane. Karyokinesis is the process of dividing a cell's nucleus to from to nuclei for the two daughter cells during cell division.

Difference between gene and development:

Gene: A gene is the basic physical and functional unit of heredity. Genes are made up of DNA. Some genes act as instructions to make molecules called proteins.

Development: the progressive changes in size, shape, and function during the life of an organism by which its genetic potentials (genotype) are translated into functioning mature systems (phenotype).

Relation b/w gene and development:

Following important points depict the relationship of genes and development.

1= Development connects genotype and phenotype. 2= Nuclear genes are not lost or mutated during development. The genome of every cell is equivalent. 3= The exceptions to the rule of genomic equivalence are the lymphocytes. During differentiation, these cells rearrange their DNA to create new immunoglobulin and antigen receptor genes. 4= The ability of nuclei from differentiated cells to direct the development of complete adult organisms has recently confirmed the principle of genomic equivalence. 5= Only a small percentage of the genome is expressed in any particular cell. 6= The cloning of human beings, as well as regenerating damaged organs or enhancing physical abilities, may soon be possible through cloning technology and the use of embryonic stem cells.

Define reproduction:

Reproduction is the production of offspring. There are two main forms: sexual and asexual reproduction. In sexual reproduction, an organism combines the genetic information from each

of its parents and is genetically unique. In asexual reproduction, one parent copies itself to form a genetically identical offspring.

What are somites and what is their function?

Somites are literally the building blocks of the vertebrate body plan; they are essential for segmentation, bone and musculature development, as well as creating a template for the nervous system.